

Nicholas Tang

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[Portfolio](#) — [GitHub](#) — [LinkedIn](#)

Education

UC Santa Cruz

Sept 2024 – Jun 2027

B.S. Computer Science, EE Minor

GPA: 3.9

Relevant Coursework: Signals and Systems, Embedded Operating Systems, Computer Architecture, Data Structures

Experience

Rivian and Volkswagen Group Technologies

June 2026 – Sept 2026

Software Engineering Intern, Charging

- Extended protocol conformance for DIN 70121 EV charging communications, expanding unit and Software-in-the-Loop (SIL) test suites to permanently validate and lock in protocol compliance.
- Automated Hardware-in-the-Loop (HIL) integration tests in Python on a general-purpose communications test bench to evaluate cross-brand Rivian and VW vehicle software.
- Configured and implemented 2 new Unified Diagnostic Services (UDS) Data Identifiers (DIDs) on charging ECUs to enforce State of Charge (SoC) limits and enable dynamic pluglock control.

Formula Slug (FSAE) @ UC Santa Cruz

Sept 2024 – Present

Firmware Engineer

- Implemented real-time CAN-bus lap tracking firmware in C++ using flat-Earth position approximation, maintaining spatial accuracy within a ± 10 -meter margin across thousands of telemetry data points.
- Integrated VN-200 IMU/GPS module over a high-throughput 10 MHz UART interface for deterministic vehicle dynamics data acquisition.
- Developed STM32 MbedOS firmware for closed-loop PID traction control and regenerative braking energy management to improve vehicle handling stability and energy efficiency.
- Engineered safety-critical low-voltage protection routines featuring rapid undervoltage detection and automated relay actuation to protect battery cell integrity.

UCSC Baskin Engineering (ECE)

Mar 2026 – Present

Undergraduate Researcher, High Performance Power Electronics Lab

- Designed a multi-layer breakout PCB in Altium Designer for the TI C2000 microcontroller family to accelerate lab prototyping.
- Integrated dual synchronous buck converters ($5V \rightarrow 3.3V$ and $3.3V \rightarrow 1.2V$) to deliver stable rail voltages for MCU core and I/O.
- Executed PCB layout routing, trace optimization, ground plane design, and bill of materials (BOM) sourcing.

UCSC Earth and Planetary Sciences

Jun 2025 – Jan 2026

Undergraduate Researcher, Planetary Cloud Tracking

- Developed parallelized C/C++ computer vision pipelines implementing optical flow algorithms to extract planetary atmospheric wind vectors from satellite imagery.
- Built pipelines to parse, transform, and normalize large-scale spatial datasets across HDF5, NetCDF, and TIFF formats.
- Formulated numerical integrators to solve continuous Euler-Lagrange fluid equations to model atmospheric motion.

Projects & Other Experience

NASA's Professional Development Program (NPWEE)

June 2025 – Aug 2025

Lead Systems Engineer

- Led systems architecture design and co-authored a technical proposal for mid-air battery swap infrastructure in electric aviation.
- Conducted trade studies on airspace integration, automated power transfer docking mechanisms, and operational safety protocols.

Musical Auto-Transcribe DSP

Dec 2025 – Present

Personal Project

- Engineered a C-based digital signal processing pipeline that converts raw audio into human-readable musical notation and MIDI output.
- Applied Fast Fourier Transforms (FFT), Hann windowing, and custom denoising filters to perform spectral analysis and pitch detection on complex audio signals.

Skills

Languages:

C, C++, Python, Bash

Embedded & HW:

RTOS (MbedOS), STM32, TI C2000, CAN bus, UART, SPI, UDS Diagnostics, HIL/SIL Testing

EDA & Tools:

Altium Designer, KiCad, Siemens NX, Oscilloscopes, CMake, Ninja, Docker, Valgrind, GDB, Git